

Question 1 Random Forest: Full Trait Model

This notebook evaluates the custom CSC 466 random forest using `data/processed/breed_forest_full.csv`.

This version keeps each AKC trait rating as its own predictor, while `Coat Type` and `Coat Length` remain categorical.

Final Test Results

The model is trained on the training split using the best cross-validation parameters, then evaluated on the held-out test split. Macro F1 is useful here because the three popularity tiers are not perfectly balanced.

Model Interpretation

These tables show which features the forest split on most often and how average feature values differ across the actual popularity tiers.

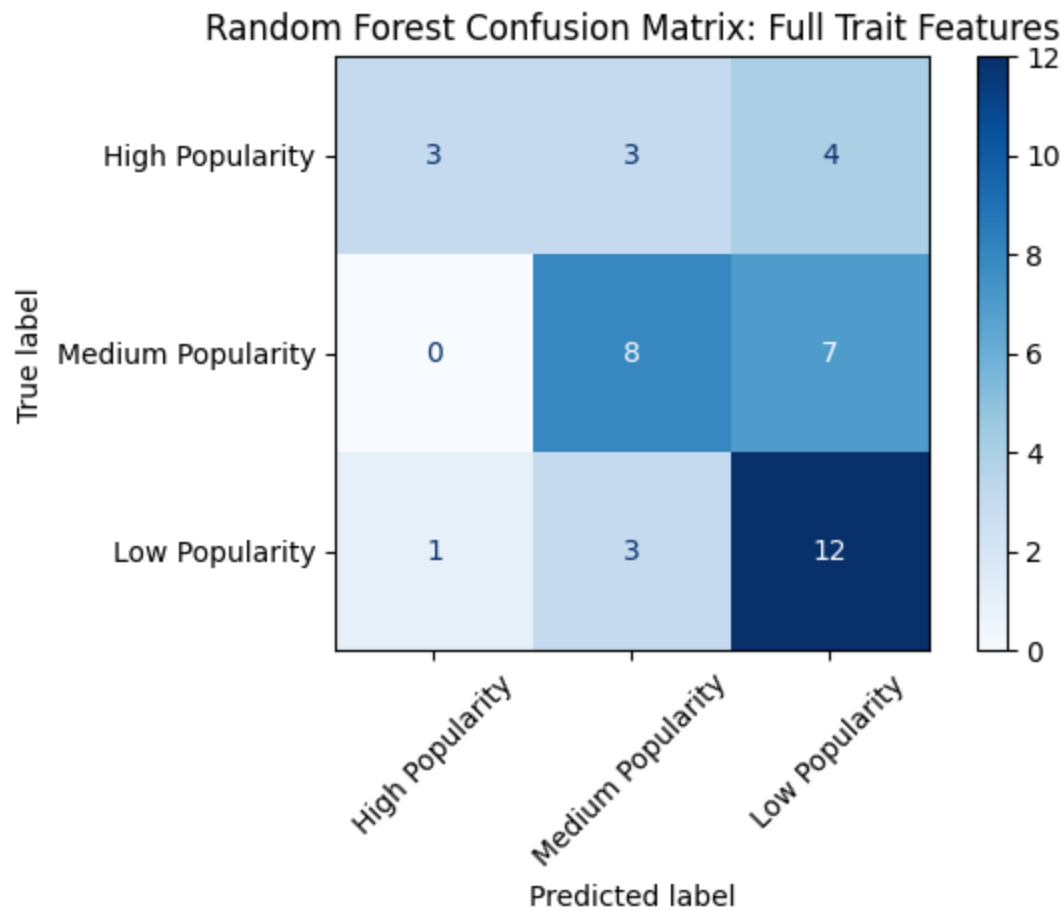
	Feature	Split Count
0	Barking Level	45
1	Watchdog/Protective Nature	39
2	Good With Other Dogs	32
3	Coat Grooming Frequency	28
4	Openness To Strangers	27
5	Shedding Level	26
6	Trainability Level	22
7	Energy Level	20
8	Affectionate With Family	17
9	Mental Stimulation Needs	16
10	Adaptability Level	16
11	Good With Young Children	12
12	Drooling Level	12
13	Playfulness Level	12

Popularity_Tier	High Popularity	Medium Popularity	Low Popularity
Affectionate With Family	4.65	4.46	4.47
Good With Young Children	4.37	3.70	3.73
Good With Other Dogs	3.82	3.29	3.62
Shedding Level	2.78	2.62	2.44
Coat Grooming Frequency	2.49	2.30	2.09
Drooling Level	1.90	1.88	1.58
Openness To Strangers	3.76	3.55	3.25
Playfulness Level	4.00	3.64	3.44
Watchdog/Protective Nature	3.76	3.92	3.49
Adaptability Level	4.08	3.71	3.66
Trainability Level	4.14	3.76	3.78
Energy Level	3.84	3.72	3.71
Barking Level	3.18	2.95	3.31
Mental Stimulation Needs	3.84	3.66	3.58

Popularity_Tier	Coat Type	Coat Length
High Popularity	Smooth	Short
Medium Popularity	Smooth	Medium
Low Popularity	Smooth	Short

	precision	recall	f1-score	support
High Popularity	0.75	0.30	0.43	10
Medium Popularity	0.57	0.53	0.55	15
Low Popularity	0.52	0.75	0.62	16
accuracy			0.56	41
macro avg	0.61	0.53	0.53	41
weighted avg	0.60	0.56	0.55	41

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<sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x147bd0590>
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Prediction CSV

This output compares each test-set breed's actual popularity tier to the tier predicted by the full-trait random forest model. **Average Rank** is included for interpretation only; it is not used as a predictor.

	Breed	Average Rank	Actual Popularity Tier	Predicted Popularity Tier	Correct
0	Doberman Pinscher	15.5	High Popularity	Medium Popularity	False
1	Dogue de Bordeaux	73.2	Medium Popularity	Medium Popularity	True
2	Sussex Spaniel	174.3	Low Popularity	Low Popularity	True
3	Australian Cattle Dog	53.1	Medium Popularity	Low Popularity	False
4	Maltese	34.8	High Popularity	Low Popularity	False
5	Silky Terrier	108.5	Medium Popularity	Low Popularity	False
6	Yorkshire Terrier	10.4	High Popularity	High Popularity	True
7	Chinese Shar-Pei	65.5	Medium Popularity	Medium Popularity	True
8	Lakeland Terrier	145.5	Low Popularity	Low Popularity	True

9	Akita	50.6	Medium Popularity	Medium Popularity	True
10	Irish Wolfhound	75.8	Medium Popularity	Low Popularity	False
11	Bloodhound	51.2	Medium Popularity	Low Popularity	False
12	English Springer Spaniel	26.9	High Popularity	Medium Popularity	False
13	Greater Swiss Mountain Dog	77.7	Medium Popularity	Medium Popularity	True
14	Petit Basset Griffon Vendéen	154.0	Low Popularity	Low Popularity	True
15	Greyhound	142.8	Low Popularity	Low Popularity	True
16	American Eskimo Dog	120.4	Medium Popularity	Medium Popularity	True
17	Boston Terrier	22.2	High Popularity	Low Popularity	False
18	Pug	32.9	High Popularity	High Popularity	True
19	Harrier	186.6	Low Popularity	Low Popularity	True
20	German Wirehaired Pointer	63.2	Medium Popularity	Low Popularity	False
21	Afghan Hound	108.9	Medium Popularity	Low Popularity	False
22	German Shepherd Dog	2.8	High Popularity	High Popularity	True
23	American Water Spaniel	162.8	Low Popularity	Low Popularity	True
24	Pumi	157.9	Low Popularity	Low Popularity	True
25	Manchester Terrier	130.2	Low Popularity	Low Popularity	True
26	Azawakh	192.4	Low Popularity	Low Popularity	True
27	Neapolitan Mastiff	110.1	Medium Popularity	Medium Popularity	True
28	Norfolk Terrier	130.8	Low Popularity	Medium Popularity	False
29	Welsh Springer Spaniel	130.2	Low Popularity	Low Popularity	True
30	Chihuahua	30.2	High Popularity	Low Popularity	False
31	Bulldog	5.8	High Popularity	Low Popularity	False
32	Bearded Collie	130.3	Low Popularity	High Popularity	False
33	Shiba Inu	44.3	High Popularity	Medium Popularity	False
34	Redbone Coonhound	144.5	Low Popularity	Low Popularity	True
35	American English Coonhound	173.1	Low Popularity	Low Popularity	True
36	Otterhound	180.2	Low Popularity	Medium Popularity	False
37	Black Russian Terrier	124.6	Medium Popularity	Medium Popularity	True
38	Beauceron	125.5	Low Popularity	Medium Popularity	False
39	Schipperke	110.4	Medium Popularity	Low Popularity	False
40	Leonberger	99.0	Medium Popularity	Medium Popularity	True